Cloud Workload Forecasting Report

1. Executive Summary

This Azure data-v1 sample is useful for short-horizon forecasts, but too short to establish how predictable cloud demand is in general. It contains **10,000 VMs** that report throughout the window, with mean training CPU of **9.4141%**. That average hides heavily utilized VMs and does not measure capacity-weighted utilization or absolute compute demand.

Persistence remains selected for both targets. Original training-only backtests give CPU MAE **0.235310 percentage points (pp)**, RMSE **0.274629 pp**, bias **+0.058641 pp**, and peak MAE **0.306193 pp**. MAE is **2.4995% of the training CPU mean**, not pointwise percentage error. All seven count candidates have zero error because population never changes, not because they predict churn.

Other training-only schedules favor a trailing mean or damped trend without changing selection. The final 24 target observations are never scored or plotted as forecast outcomes. Use persistence with an illustrative **0.50 pp buffer**, segment checks, and operator review, not automatic scaling or guaranteed coverage.

2. Data Understanding And Preprocessing

The audit records **1,340,000 readings**, 10,000 matching metadata rows, and 134 timestamps from trace-relative second 0 to 39,900 at exact 300-second cadence. Each VM has 134 readings. It found no missing fields, duplicate keys, invalid CPU values, metadata mismatches, or cadence irregularities. This yields only **134 aggregate timestamps, 110 for development**, not 1.34 million independent forecasting examples.

Cleaning requires nonblank VM IDs and finite CPU percentages in [0,100], with min_cpu <= avg_cpu <= max_cpu. Values in [0,1] are not rescaled. Invalid rows are excluded with warnings and audit counts. Valid duplicates are averaged within VM/timestamp to prevent duplicate weighting. Missing readings contribute neither VM nor CPU; missing or recordless timestamps and irregular cadence raise errors. No imputation or resampling occurs.

Before the chronological **110/24 split**, active_vm_count counts distinct represented VMs; aggregate_cpu_utilization averages their avg_cpu on the full grid. Neither estimates unobserved fleet members. CPU EDA uses only the first 110 timestamps; full-window coverage checks are structural.

All recorded creation times are zero; deletion-minus-creation spans have minimum 40,500 and median 2,591,400 seconds. These **recorded metadata spans** verify neither uncensored lifetimes nor simultaneous creation. Inclusive endpoints are an audit convention; source censoring/boundary semantics remain uncertain. No bounds violations occurred. Metadata CPU summaries have no above-100 values but unknown origin-time availability, so are excluded as predictors. Core/memory labels are categories, not exact capacities.

Complete traces in the first three source shards, category/core stratification, and stable SHA-256 ranking determine selection. This favors survivors, suppresses churn, and limits representativeness. Coverage, real creation/deletion, absolute compute demand, and equal-VM mean CPU remain distinct.

3. Workload Analysis

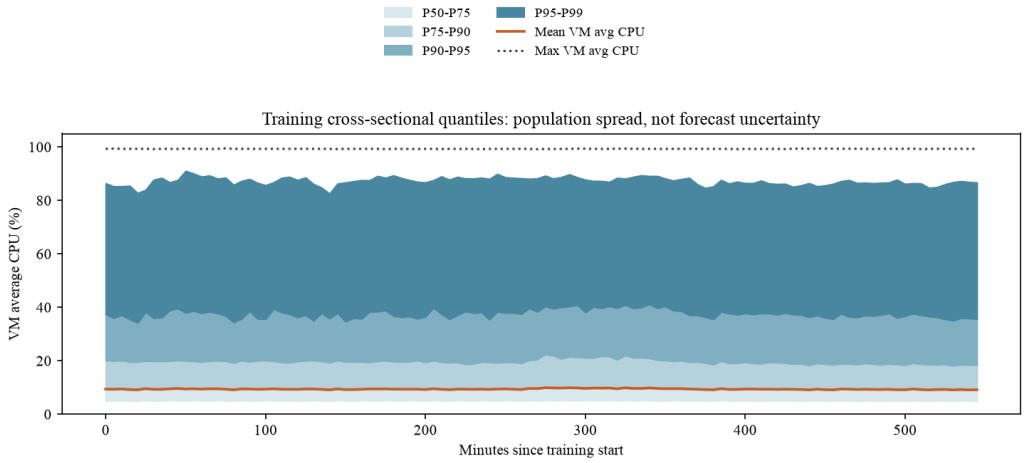


Figure 1: Training cross-sectional CPU distribution; population percentiles are not forecast uncertainty.

Training mean CPU spans **9.1116%-9.9122%** (temporal std. **0.1936 pp**), while cross-VM P95 spans **34.2264%-41.1702%** and P99 **82.5398%-91.0373%** per timestamp. The fan shows population dispersion, not forecast uncertainty. Low means can hide resource-starvation risk, but these traces establish neither starvation nor SLA violations. Lag-one autocorrelation **0.6672** and largest five-minute change **0.4368 pp** support persistence locally, not across unobserved daily/weekly cycles. Trace-relative time has no established calendar/timezone mapping.

Representation is not activity. Each timestamp represents **10,000 VMs**; the illustrative workload-active rule avg_cpu > 5 AND max_cpu > 20 counts **2,561-2,951**. Training VM means classify **4,635 low (<5%)**, **5,145 moderate (5%-50%)**, and **220 high (>50%)**. Separately, **437** VMs ever have interval-average CPU >80%, **2,235** interval-maximum CPU >80%, and **1,000** meet the top-decile within-VM variability flag (std. q90 **6.1846 pp**). These descriptive flags overlap; they are not incident labels.

Pooled readings have median **5.1198%**, P95 **37.2932%**, and P99 **87.4063%**; **1.2721%/0.5663%** reach 80%/95%. The retrospectively hottest 100 VMs contribute **71.61% of 13,993 >=80% readings**, warranting inspection of persistent hot subsets.

Delay-insensitive, Interactive, and Unknown VMs average **8.2573%, 11.5051%, and 10.3547%**; their >=80% reading shares are **0.8923%, 1.9120%, and 1.7783%**. These differences warrant separate monitoring, but workload mix is only a plausible explanation, not a causal finding.

Category-mean temporal variability differs from cross-VM dispersion:

Category	VMs	Temporal std. (pp)	Largest 5-min change (pp)
Delay-insensitive	6,180	0.2170	0.3936
Interactive	3,091	0.2433	0.7157
Unknown	729	0.6290	0.8489

Smaller groups smooth less variation; Unknown's volatility is not causal evidence. See segment variability; tiny buckets like >64 memory (10 VMs) are unrepresentative. These are not segment forecasts.

4. Forecasting Approach

Seven deterministic univariate candidates are fixed before EDA: last-value persistence; trailing means over **6, 12, or 24** observations; simple exponential smoothing (SES) with alpha **0.2 or 0.5**; and damped Holt trend with fixed alpha **0.3**, beta **0.1**, and damping **0.9**. These test smoothing and restrained trend extrapolation against the latest level.

Persistence and means forecast constants; means use available trailing history. SES starts at the first value, updates recursively, then holds its final level. Damped trend starts at zero trend and projects a geometrically damped continuation without coefficient optimization. Counts use np.rint nearest-integer rounding (ties to even), bounded below by zero; CPU is clipped to [0,100].

Models use only past targets, not deletion times, metadata CPU summaries, future aggregates, or calendar features. Seasonal-naive lacks an established period; trees and neural networks lack substantial independent history. Exclusion controls scope, not potential usefulness. The selected model refits on **110 training observations** for 24 forecasts without holdout updates.

5. Validation Methodology

Original expanding windows fit prefixes of **50, 70, and 86** observations and forecast **24 five-minute steps**, matching the final two-hour task. Zero-based windows **50-73, 70-93, and 86-109** yield **72 pooled errors** per candidate/target but only **60 distinct timestamps**. Overlapping decisions are not independent.

Selection minimizes pooled MAE per target; candidate-order ties favor persistence. Candidates share origins, horizons, and metrics. Selection freezes before training-only sensitivity checks on schedules **38/62/86** and **30/54/78**, which do not revise it.

Error is prediction minus actual. RMSE emphasizes large errors; bias gives direction. **Normalized MAE = MAE / training-target mean**, expressed as a percentage, uses a fixed training-only denominator, not near-zero individual outcomes. CPU's denominator is **9.414089019539009%**. Improvement is relative to the best persistence/mean baseline and undefined when its MAE is zero.

Peak cases have actual CPU at least the fitting prefix's **90th percentile**, recomputed from past data at each origin. Substituting extreme finite and missing holdout labels leaves predictions, backtests, and selection unchanged in the holdout isolation check. This verifies forecasting-boundary label isolation, not accuracy.

6. Results Versus Baseline

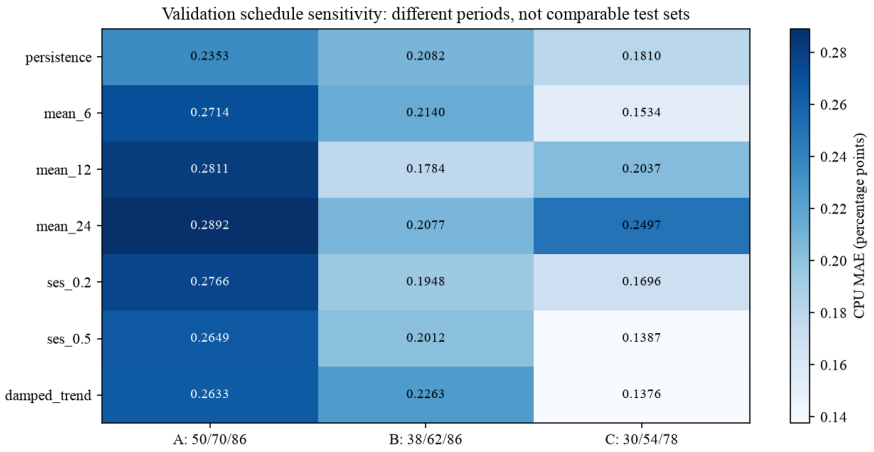


Figure 2: Seven CPU candidates across three training-only schedules; pooled MAE is schedule-sensitive.

Original-schedule errors are in **pp**; rounding leaves rankings unchanged.

Candidate	MAE	RMSE	Peak MAE
Persistence	**0.235310**	**0.274629**	**0.306193**
Mean 6	0.271409	0.314064	0.364213
Mean 12	0.281126	0.329910	0.373491
Mean 24	0.289180	0.318631	0.371043
SES 0.2	0.276596	0.320205	0.367535
SES 0.5	0.264916	0.308751	0.352657
Damped trend	0.263274	0.311696	0.363078

Persistence wins original-schedule MAE, RMSE, and peak MAE, with normalized MAE **2.499548%**. Damped trend's MAE is **11.88% worse** despite lower bias. All count candidates have MAE/RMSE/bias/peak MAE **0**, normalized MAE **0%**, and undefined baseline-relative improvement.

Origins	Diagnostic MAE favorite	Best MAE	Persistence MAE
50/70/86	Persistence	0.235310	0.235310
38/62/86	Mean 12	0.178385	0.208198
30/54/78	Damped trend	0.137625	0.181039

The sensitivity results show **ranking instability**, not superiority across differently scored periods. The full schedule results retain MAE, RMSE, bias, and maximum absolute error for every candidate, by origin and pooled. Count errors remain zero; selection stays frozen.

7. Error And Failure Analysis

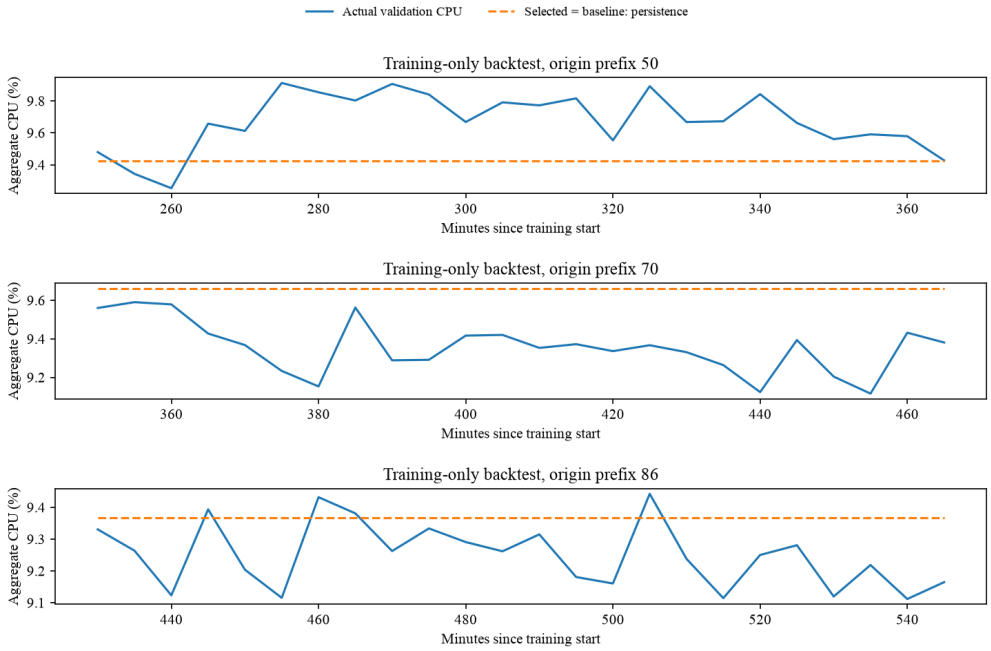


Figure 3: Three training-only origins expose opposite persistence failure modes; selected model equals baseline.

Origin errors are in **pp**. Missing peak MAE means no qualifying cases, not zero error.

Origin	MAE	RMSE	Bias	Max abs.	Peaks	Peak MAE
50	0.2681	0.3028	-0.2467	0.4858	20	0.3062
70	0.3054	0.3329	+0.3054	0.5468	0	N/A
86	0.1324	0.1542	+0.1172	0.2551	0	N/A

Persistence misses a rise at origin 50 and overpredicts the lower level at origin 70. Pooled bias **+0.058641 pp** masks opposing risks; it offers no reliable peak protection. Maximum absolute error **0.5468 pp** is an overprediction; maximum underprediction is **0.48582143 pp**.

All **20 peak cases** occur at origin 50, with past-only threshold **9.5202%**. Origins 70 and 86 have thresholds **9.8038%** and **9.7815%**, with no qualifying outcomes. Pooled peak MAE describes one regime, not three independent peak tests.

Pooled first-six-step MAE is **0.1806 pp**, versus **0.2491 pp** for the last six. By origin, first/last-six MAEs are **0.2020/0.1840**, **0.2024/0.3873**, and **0.1373/0.1759 pp**. Error is not monotonic: origin 50 improves late; origin 70 deteriorates. Three decisions per horizon make estimates noisy; origin/horizon errors support inspection, not confidence intervals. Refreshing forecasts with new readings needs separate validation.

Underprediction occurs in **22/24**, **0/24**, and **4/24** cases by origin. Retrospectively, fitting-prefix positive-change q90 defines large upward steps: **2** qualify versus **26** smaller steps. Their persistence MAEs, **0.1654/0.2263 pp**, do **not** establish greater vulnerability to abrupt rises. Smaller steps need not be gradual trends; noisy groups confound origin and horizon. All-candidate downside and change-group tables remain in the appendix.

8. Capacity-Planning Recommendation

Use persistence for monitoring, not capacity allocation. Underprediction magnitude is max(actual - prediction, 0); selected-validation q95 is **0.42116067 pp**, maximum **0.48582143 pp**. Rounding the maximum up gives the unchanged illustrative **0.50 pp buffer**, not calibrated coverage: these overlapping errors also served selection.

Separately, the horizon schedule adds an illustrative allowance to the frozen **9.164556%** forecast: P95-envelope allowance **0.047577 pp at 5 minutes** versus **0.437239 pp at 120 minutes**, yielding mean-CPU references **9.212133%/9.601795%**. These cumulative maxima of raw positive-part residual quantiles use only **three samples per horizon**. They are not coverage probabilities, new predictions, buffer retuning, or absolute capacity requirements. The appendix distinguishes raw P90/P95 from the conservative monotone display heuristic.

Compare min(100, forecast + 0.50) with a **team-configured threshold T** for review, not machine additions. Individual 80%/95% cutoffs are not aggregate alert thresholds. Larger buffers reduce missed rises but increase alerts and possible overprovisioning; set thresholds from independent service objectives, scaling latency, and incident costs.

Hypothetical, not an Atomity threshold: forecast 9.6% + 0.50 pp = 10.1% crosses T = 10.0%, prompting segment CPU, latency/error-rate, and coverage checks. Underprediction could leave resources short during scaling delay; overprediction could waste resources or review effort. Asymmetric loss weighting remains hypothetical and service-dependent: no observed SLA outcomes or cost ratio establish either as far costlier.

Monitor high-utilization segments alongside the mean. Coverage loss, delayed telemetry, buffer exceedances, or rising segment high-load shares warrant review, but are not validated incident detectors. Do not downsize on a low mean: without capacities and placement, equal-VM CPU cannot determine cores, VM additions, or absolute demand.

9. Limitations And Future Work

Suppressed churn prevents transferring count accuracy to arrivals/departures. The two-hour horizon is **21.8% of 110 training points**; local levels, ramps, and reversals matter. Overlap and selection weaken evidence; peaks cover one origin. Results establish no external generalization, causal segment effect, calibrated uncertainty, or holdout performance.

Collect at least **seven days (2,016 five-minute samples)** before testing future candidates: an equal-weight persistence/Mean-12 ensemble or adaptive window. Seven days is a collection milestone, not sufficient evidence or proven improvement. Before production changes, reserve non-overlapping evaluation, validate buffers independently, measure service losses, and use origin-available metadata and trustworthy capacities.

10. Reproducibility And Artifacts

The README covers release checksums, installation, tests, prediction, and analysis commands. The executed notebook uses the tested package. Provenance resides in results/selection.json, audit.json, and analysis_summary.json; exclusions in the decision log. Root predictions.csv and predictions.selection.json preserve unscored final forecasts and selection. scripts/render_report.py renders this source; scripts/verify_clean_copy.py checks isolated reproduction.

Appendix: Supporting Evidence

Main figures appear inline only. Supporting analysis is training-only; full-window data-quality coverage is structural, not a holdout forecast outcome.

One-sided loss is max(actual - prediction, 0), including zeros for overpredictions. The directional comparison summarizes all seven candidates identically, by original origin and pooled. Persistence's pooled underprediction frequency is **26/72 (36.11%)**, zero-inclusive mean shortfall **0.088335 pp**, q95 **0.421161 pp**, and maximum **0.485821 pp**. This describes observed downside, not independent calibration or grounds to retune selection.

Origin	Underprediction cases	Mean shortfall (pp)	Q95 shortfall (pp)
50	22/24	0.257413	0.477701
70	0/24	0.000000	0.000000
86	4/24	0.007590	0.059398

Change diagnostics and step records retain all candidates, N/A empty groups, and past-only cutoffs. Actual step labels serve validation only, never forecasting, selection, or operational threshold calibration.

Horizon headroom construction. Residuals here are actual - prediction, opposite to stored signed error = prediction - actual. Each horizon has three positive-part samples, including overprediction zeros, from overlapping original validation windows. Raw P90/P95 use empirical order statistics with linear interpolation, not calibrated 90%/95% future coverage. Cumulative maxima impose monotonicity for conservative display; raw empirical risk need not increase. Adding each envelope to the frozen forecast and clipping to [0,100] gives a mean-CPU reference, not provisioned compute. The 24-row schedule retains raw quantiles, envelopes, and references; the fixed 0.50 pp review heuristic remains unchanged.

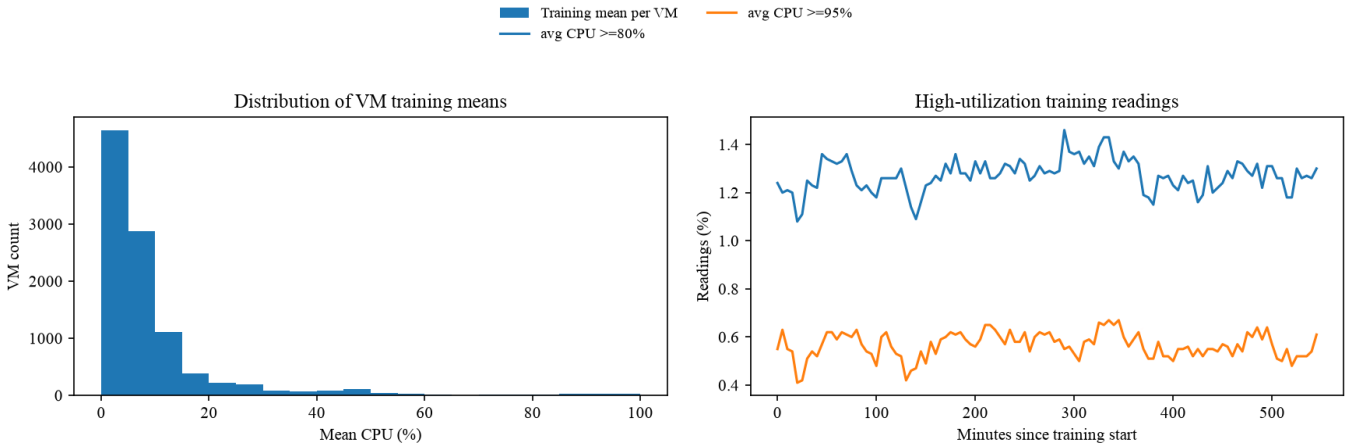
Activity and population caveats. Workload-active status combines interval average and maximum, not metadata membership. VM mean classes and within-VM standard deviations are training-only. High mean, peak average, peak interval maximum, and burstiness flags overlap; never sum them as distinct VMs. The fan shows cross-VM dispersion, not future fleet-mean uncertainty.

- VM distribution and high-utilization shares: heterogeneous utilization warrants tail monitoring. Distribution table.
- Training metadata segments: monitor differences; treat small groups cautiously. Segment table.
- Autocorrelation: supports local persistence, not seasonality. Lag table.
- Horizon errors: three origins cannot establish smooth error growth. Horizon metrics.
- Residual and peak diagnostics: bias masks directional failures and concentrated peaks. Origin metrics.
- Data quality: balanced coverage limits changing-fleet generalization.
- Training CPU and coverage: constant 10,000 represented VMs does not imply constant workload-active counts.
- Original model comparison: selection retained alongside schedule sensitivity checks.
- Horizon headroom: descriptive cumulative-max P90/P95 envelopes, not calibrated prediction intervals.

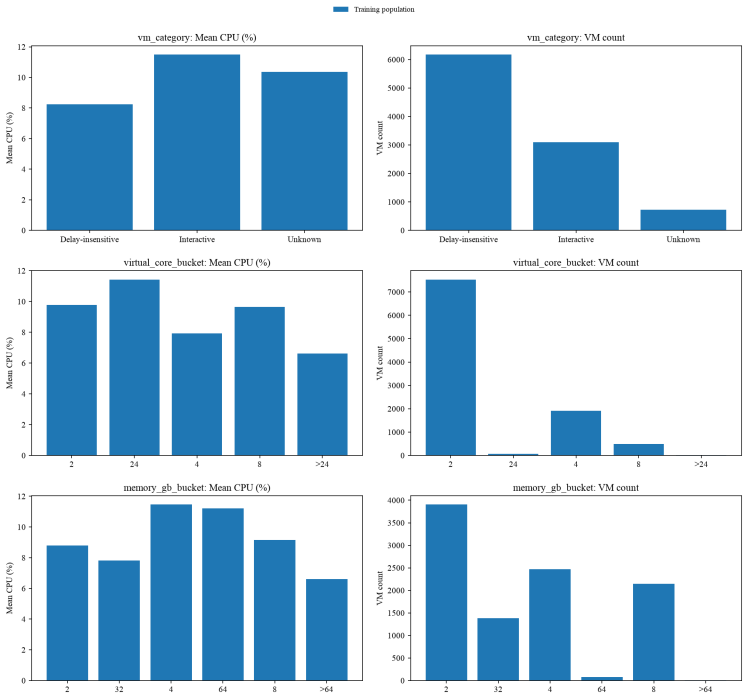
Training targets, time summaries, backtest errors, origin/horizon errors, sensitivity, missingness, and coverage permit inspection without exported holdout CPU labels.

Appendix: Figures

VM distribution and high-utilization shares

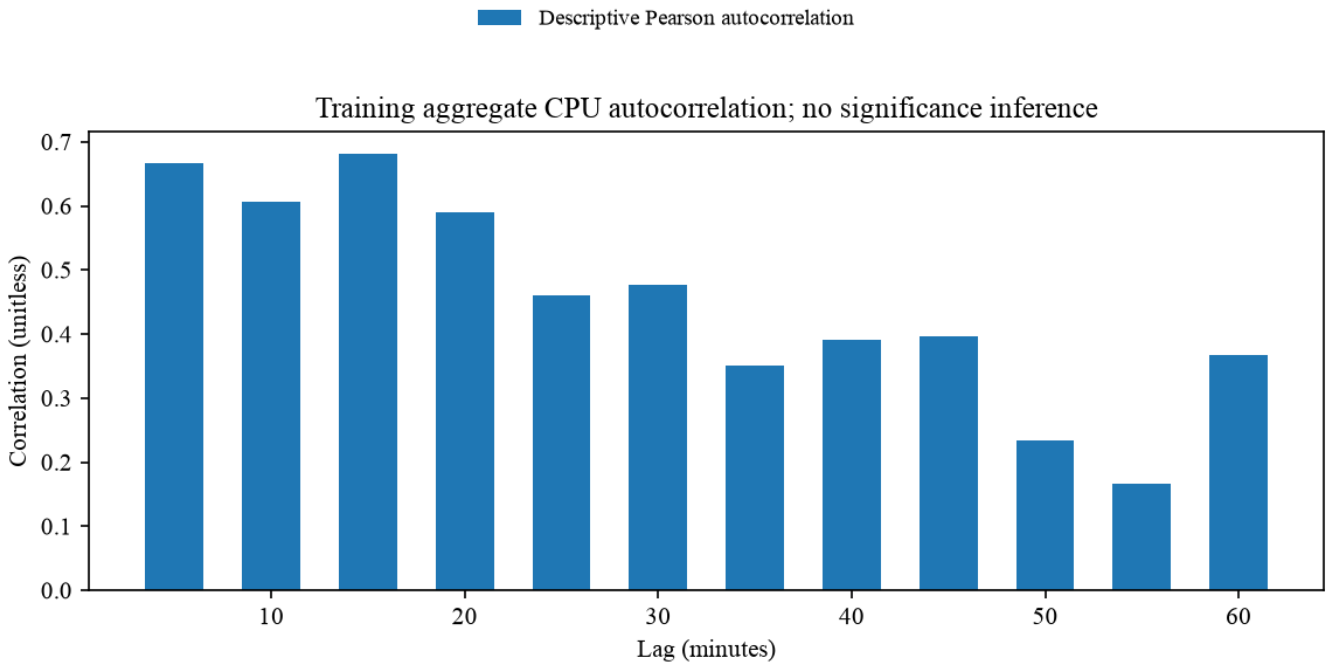


Training metadata segments

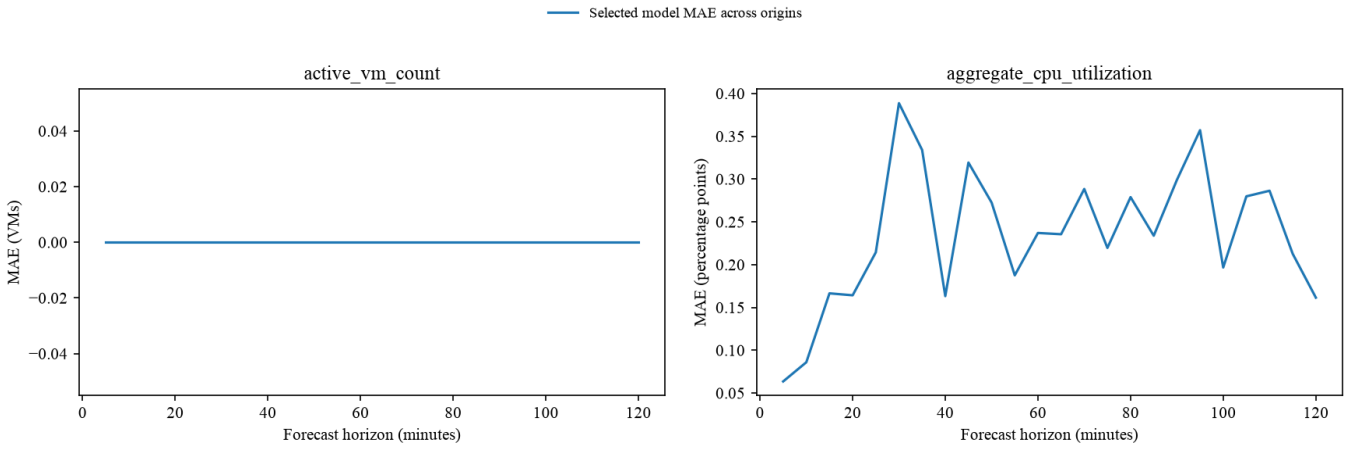


Appendix: Figures

Autocorrelation

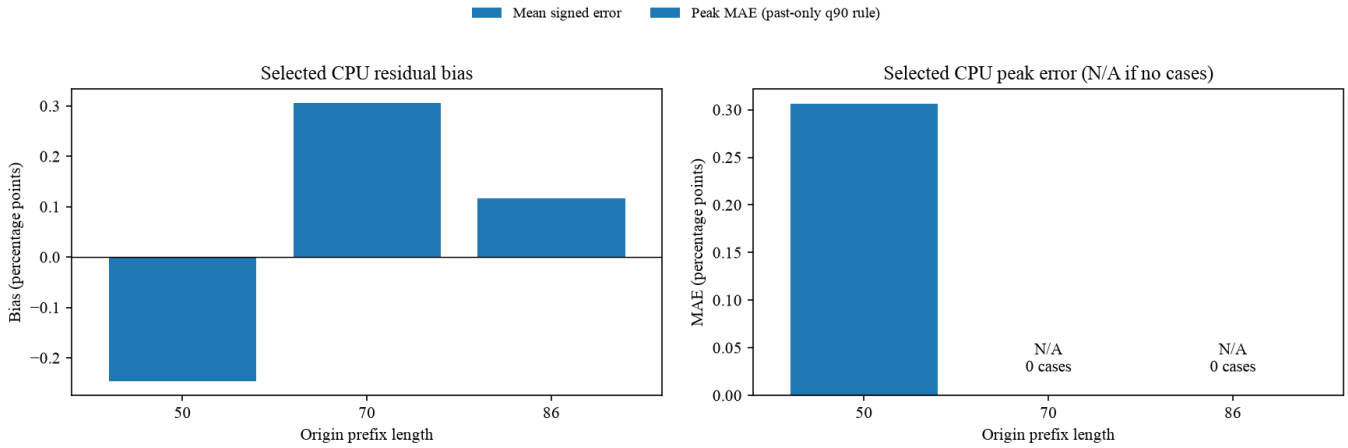


Horizon errors

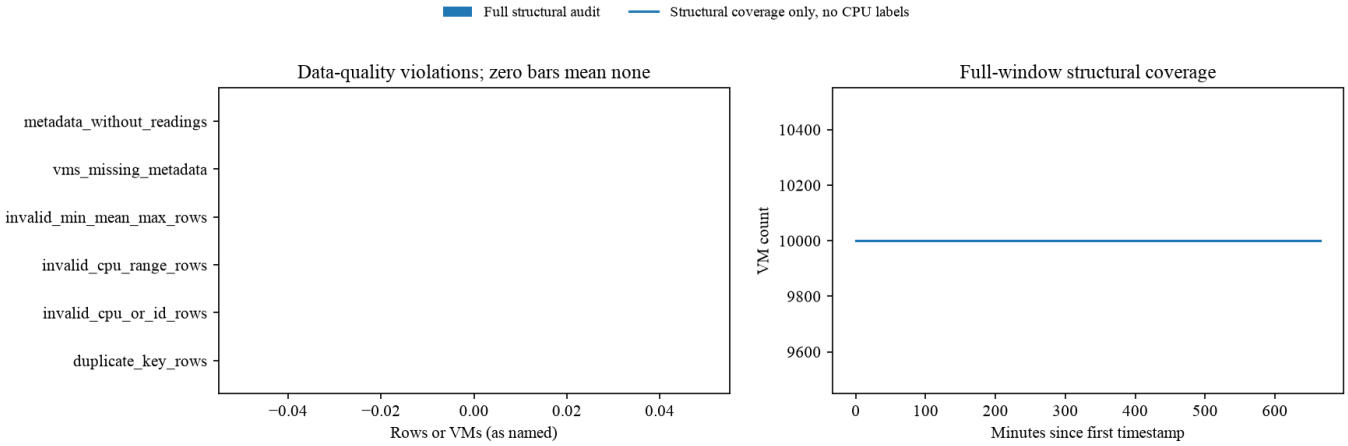


Appendix: Figures

Residual and peak diagnostics

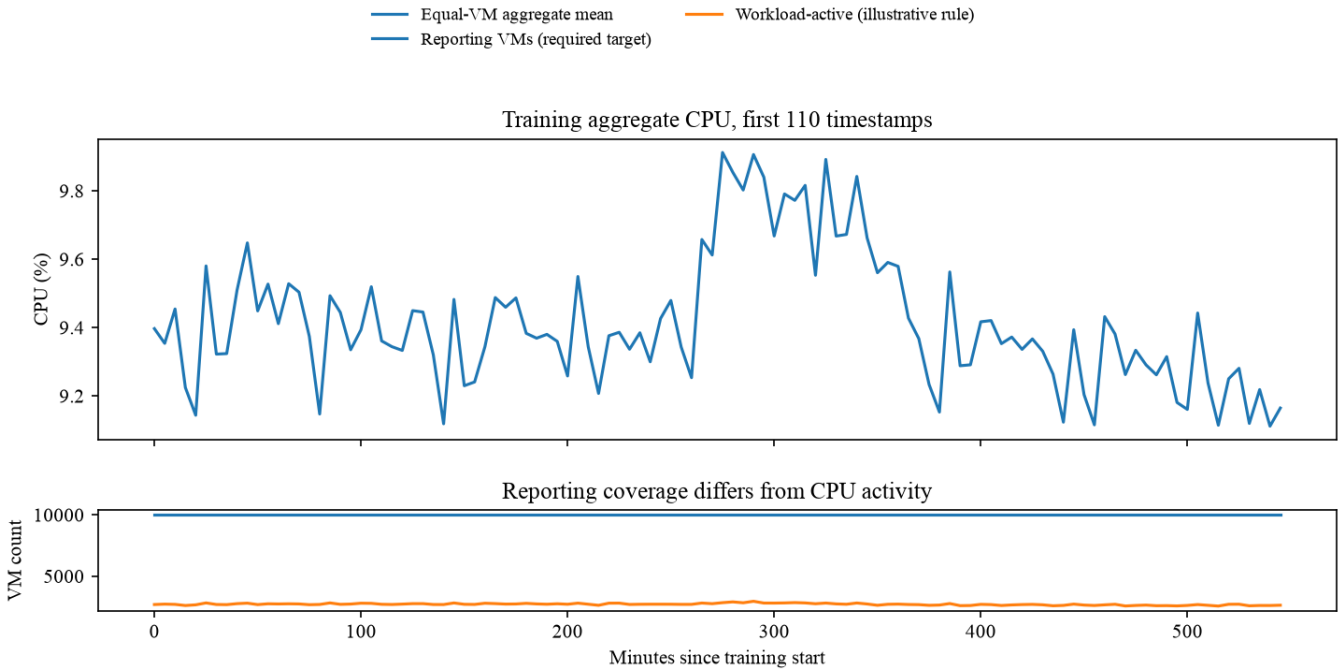


Data quality

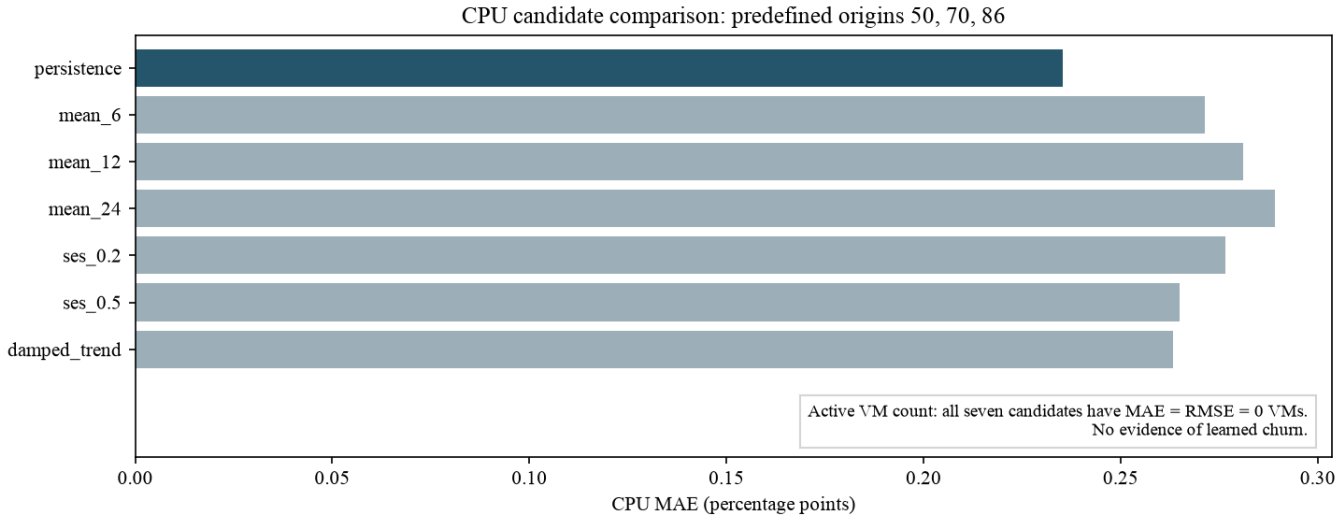


Appendix: Figures

Training CPU and coverage



Original model comparison



Appendix: Figures

Horizon headroom

